STA ANALYSIS - SKY130 HD

Post-Layout Static Timing Analysis of the Traffic Signal Controller (traffic_ctrl)

Roll No.: 24EG104E28

Execution

OpenSTA was run on my Ubuntu/UTM environment using the SKY130 HD timing libraries, on the routed design produced by OpenLane (run1). The gate-level netlist (including clock-tree buffers and physical tap/filler cells) was combined with the extracted parasitics (SPEF) so that real wire delays are included. All 16 corners were analyzed with a 50 ns clock (20 MHz). The clock is propagated through the synthesized clock tree. Clock uncertainty is 0.25 ns, input and output delays are 10 ns (20% of the period), input transition is 0.2 ns and output load is 0.05 pF. Magic reported 0 DRC errors on the final layout.

Important commands used

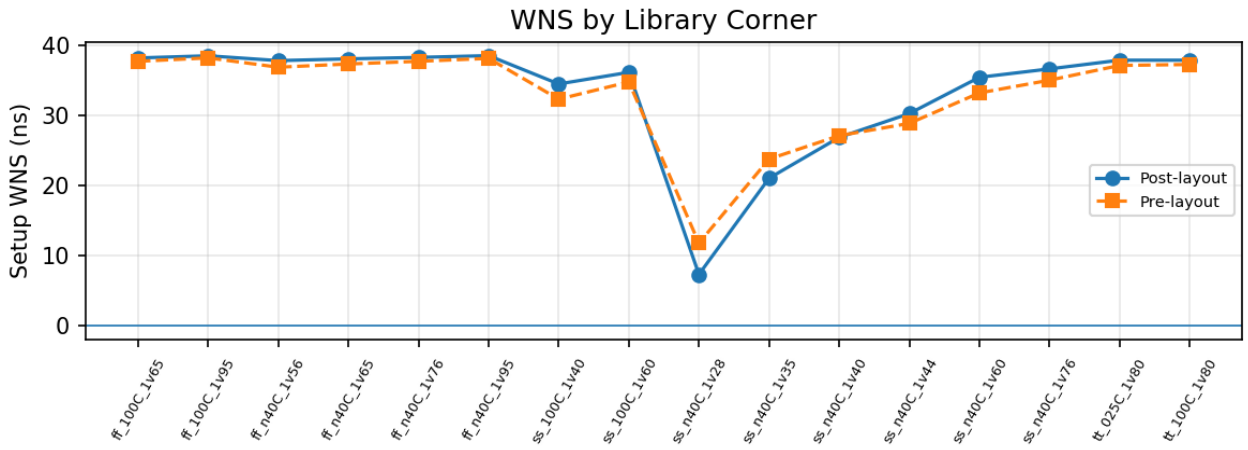
```
read_liberty <corner_library.lib>
read_verilog traffic_ctrl.synthesis_preroute.v
link_design traffic_ctrl
read_sdc post.sdc
read_spef traffic_ctrl.spef
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}
```

Notes

The SDC differs from the pre-layout run in two ways: the clock is propagated (set_propagated_clock) instead of ideal, and no fixed clock transition is set, because the real clock-tree slew is used. OpenSTA also prints "module not found" warnings for tap and filler cells; these are physical-only cells with no timing arcs, so they do not affect the results.

16-Corner Result Summary (Post-Layout, SPEF)

#	Library corner	Process / Temp / V	Hold slack (ns)	Setup slack (ns)	Status
1	ff_100C_1v65	FF / +100 C / 1.65 V	0.12	38.21	MET
2	ff_100C_1v95	FF / +100 C / 1.95 V	0.05	38.52	MET
3	ff_n40C_1v56	FF / -40 C / 1.56 V	0.19	37.83	MET
4	ff_n40C_1v65	FF / -40 C / 1.65 V	0.14	38.08	MET
5	ff_n40C_1v76	FF / -40 C / 1.76 V	0.09	38.29	MET
6	ff_n40C_1v95	FF / -40 C / 1.95 V	0.04	38.54	MET
7	ss_100C_1v40	SS / +100 C / 1.40 V	1.06	34.50	MET
8	ss_100C_1v60	SS / +100 C / 1.60 V	0.67	36.16	MET
9	ss_n40C_1v28	SS / -40 C / 1.28 V	2.34	7.35	MET
10	ss_n40C_1v35	SS / -40 C / 1.35 V	1.70	21.06	MET
11	ss_n40C_1v40	SS / -40 C / 1.40 V	1.39	26.88	MET
12	ss_n40C_1v44	SS / -40 C / 1.44 V	1.20	30.28	MET
13	ss_n40C_1v60	SS / -40 C / 1.60 V	0.73	35.45	MET
14	ss_n40C_1v76	SS / -40 C / 1.76 V	0.47	36.65	MET
15	tt_025C_1v80	TT / +25 C / 1.80 V	0.21	37.89	MET
16	tt_100C_1v80	TT / +100 C / 1.80 V	0.21	37.89	MET



Negative setup slack indicates a setup timing violation for the reported worst max path.

Pre-Layout vs Post-Layout Comparison

Library corner	Setup pre (ns)	Setup post (ns)	Hold pre (ns)	Hold post (ns)
ff_100C_1v65	37.73	38.21	0.12	0.12
ff_100C_1v95	38.22	38.52	0.04	0.05
ff_n40C_1v56	36.89	37.83	0.18	0.19
ff_n40C_1v65	37.35	38.08	0.13	0.14
ff_n40C_1v76	37.73	38.29	0.09	0.09
ff_n40C_1v95	38.15	38.54	0.04	0.04
ss_100C_1v40	32.33	34.50	0.99	1.06
ss_100C_1v60	34.79	36.16	0.64	0.67
ss_n40C_1v28	11.81	7.35	2.08	2.34
ss_n40C_1v35	23.79	21.06	1.50	1.70
ss_n40C_1v40	27.13	26.88	1.23	1.39
ss_n40C_1v44	28.88	30.28	1.06	1.20
ss_n40C_1v60	33.22	35.45	0.66	0.73
ss_n40C_1v76	35.04	36.65	0.44	0.47
tt_025C_1v80	37.14	37.89	0.20	0.21
tt_100C_1v80	37.28	37.89	0.20	0.21

Setup slack shrinks after layout because of wire delay and the clock tree. Hold slack is almost unchanged.

ff_100C_1v65

Meaning: Fast NMOS / Fast PMOS; temperature +100 C; supply 1.65 V.

Result: hold slack 0.12 ns; setup slack 38.21 ns - **MET**.

```
worst slack 38.21
worst slack 0.12
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.31    0.31   clock network delay (propagated)
                   0.16   0.00    0.31 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.03   0.27    0.59 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                   0.03   0.00    0.59 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                   0.04   0.09    0.68 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                   0.04   0.00    0.68 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.68   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.31    0.31   clock network delay (propagated)
                          0.25    0.56   clock uncertainty
                          0.00    0.56   clock reconvergence pessimism
                                  0.56 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.55   library hold time
                                  0.55   data required time
---------------------------------------------------------------------------
                                  0.55   data required time
                                 -0.68   data arrival time
---------------------------------------------------------------------------
                                  0.12   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.32    0.32   clock network delay (propagated)
                   0.10   0.00    0.32 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.04   0.27    0.60 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01                           state[3] (net)
                   0.04   0.00    0.60 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                   0.06   0.11    0.71 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                   0.06   0.00    0.71 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                   0.06   0.12    0.83 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                   0.06   0.00    0.83 v _209_/B (sky130_fd_sc_hd__or4_4)
                   0.08   0.40    1.22 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                   0.08   0.00    1.22 v _295_/A (sky130_fd_sc_hd__inv_2)
                   0.05   0.06    1.29 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                   0.05   0.00    1.29 ^ _297_/B (sky130_fd_sc_hd__nor2_1)
                   0.04   0.04    1.33 v _297_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.01                           net9 (net)
                   0.04   0.00    1.33 v output9/A (sky130_fd_sc_hd__clkbuf_2)
                   0.19   0.21    1.54 v output9/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_red (net)
                   0.19   0.00    1.54 v a_red (out)
                                  1.54   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.54   data arrival time
---------------------------------------------------------------------------
                                 38.21   slack (MET)
```

ff_100C_1v95

Meaning: Fast NMOS / Fast PMOS; temperature +100 C; supply 1.95 V.

Result: hold slack 0.05 ns; setup slack 38.52 ns - **MET**.

```
worst slack 38.52
worst slack 0.05
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.26    0.26   clock network delay (propagated)
                   0.14   0.00    0.26 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.03   0.21    0.47 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                   0.03   0.00    0.47 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                   0.03   0.07    0.55 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                   0.03   0.00    0.55 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.55   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.26    0.26   clock network delay (propagated)
                          0.25    0.51   clock uncertainty
                          0.00    0.51   clock reconvergence pessimism
                                  0.51 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.50   library hold time
                                  0.50   data required time
---------------------------------------------------------------------------
                                  0.50   data required time
                                 -0.55   data arrival time
---------------------------------------------------------------------------
                                  0.05   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.27    0.27   clock network delay (propagated)
                   0.09   0.00    0.27 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.03   0.22    0.48 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01                           state[3] (net)
                   0.03   0.00    0.48 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                   0.05   0.09    0.58 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                   0.05   0.00    0.58 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                   0.05   0.10    0.67 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                   0.05   0.00    0.67 v _209_/B (sky130_fd_sc_hd__or4_4)
                   0.07   0.29    0.97 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                   0.07   0.00    0.97 v _295_/A (sky130_fd_sc_hd__inv_2)
                   0.04   0.05    1.02 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                   0.04   0.00    1.02 ^ _297_/B (sky130_fd_sc_hd__nor2_1)
                   0.03   0.04    1.05 v _297_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.01                           net9 (net)
                   0.03   0.00    1.05 v output9/A (sky130_fd_sc_hd__clkbuf_2)
                   0.16   0.17    1.23 v output9/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_red (net)
                   0.16   0.00    1.23 v a_red (out)
                                  1.23   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.23   data arrival time
---------------------------------------------------------------------------
                                 38.52   slack (MET)
```

ff_n40C_1v56

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.56 V.

Result: hold slack 0.19 ns; setup slack 37.83 ns - **MET**.

```
worst slack 37.83
worst slack 0.19
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.37    0.37   clock network delay (propagated)
                  0.20    0.00    0.37 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.34    0.71 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.04    0.00    0.71 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                  0.04    0.10    0.81 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                  0.04    0.00    0.81 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.81   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.37    0.37   clock network delay (propagated)
                          0.25    0.62   clock uncertainty
                          0.00    0.62   clock reconvergence pessimism
                                  0.62 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                          0.00    0.62   library hold time
                                  0.62   data required time
---------------------------------------------------------------------------
                                  0.62   data required time
                                 -0.81   data arrival time
---------------------------------------------------------------------------
                                  0.19   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_yellow (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.39    0.39   clock network delay (propagated)
                  0.13    0.00    0.39 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.32    0.71 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           state[3] (net)
                  0.03    0.00    0.71 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.05    0.12    0.83 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                  0.05    0.00    0.83 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.05    0.13    0.96 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                  0.05    0.00    0.96 v _209_/B (sky130_fd_sc_hd__or4_4)
                  0.10    0.58    1.54 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                  0.10    0.00    1.54 v _295_/A (sky130_fd_sc_hd__inv_2)
                  0.06    0.09    1.64 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                  0.06    0.00    1.64 ^ output10/A (sky130_fd_sc_hd__clkbuf_2)
                  0.31    0.29    1.92 ^ output10/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_yellow (net)
                  0.31    0.00    1.92 ^ a_yellow (out)
                                  1.92   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.92   data arrival time
---------------------------------------------------------------------------
                                 37.83   slack (MET)
```

ff_n40C_1v65

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.65 V.

Result: hold slack 0.14 ns; setup slack 38.08 ns - **MET**.

```
worst slack 38.08
worst slack 0.14
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.33    0.33   clock network delay (propagated)
                  0.18    0.00    0.33 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.29    0.62 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.62 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                  0.04    0.09    0.71 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                  0.04    0.00    0.71 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.71   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.33    0.33   clock network delay (propagated)
                          0.25    0.58   clock uncertainty
                          0.00    0.58   clock reconvergence pessimism
                                  0.58 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.58   library hold time
                                  0.58   data required time
---------------------------------------------------------------------------
                                  0.58   data required time
                                 -0.71   data arrival time
---------------------------------------------------------------------------
                                  0.14   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_yellow (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.35    0.35   clock network delay (propagated)
                  0.12    0.00    0.35 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.28    0.63 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           state[3] (net)
                  0.03    0.00    0.63 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.05    0.11    0.74 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                  0.05    0.00    0.74 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.05    0.12    0.85 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                  0.05    0.00    0.85 v _209_/B (sky130_fd_sc_hd__or4_4)
                  0.09    0.48    1.34 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                  0.09    0.00    1.34 v _295_/A (sky130_fd_sc_hd__inv_2)
                  0.05    0.08    1.42 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                  0.05    0.00    1.42 ^ output10/A (sky130_fd_sc_hd__clkbuf_2)
                  0.28    0.26    1.67 ^ output10/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_yellow (net)
                  0.28    0.00    1.67 ^ a_yellow (out)
                                  1.67   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.67   data arrival time
---------------------------------------------------------------------------
                                 38.08   slack (MET)
```

ff_n40C_1v76

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.76 V.

Result: hold slack 0.09 ns; setup slack 38.29 ns - **MET**.

```
worst slack 38.29
worst slack 0.09
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.30    0.30   clock network delay (propagated)
                  0.16    0.00    0.30 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.25    0.55 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.55 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                  0.03    0.08    0.63 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                  0.03    0.00    0.63 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.63   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.30    0.30   clock network delay (propagated)
                          0.25    0.55   clock uncertainty
                          0.00    0.55   clock reconvergence pessimism
                                  0.55 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.54   library hold time
                                  0.54   data required time
---------------------------------------------------------------------------
                                  0.54   data required time
                                 -0.63   data arrival time
---------------------------------------------------------------------------
                                  0.09   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_yellow (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.31    0.31   clock network delay (propagated)
                  0.11    0.00    0.31 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.25    0.56 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           state[3] (net)
                  0.03    0.00    0.56 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.04    0.10    0.65 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                  0.04    0.00    0.65 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.04    0.11    0.76 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                  0.04    0.00    0.76 v _209_/B (sky130_fd_sc_hd__or4_4)
                  0.08    0.40    1.16 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                  0.08    0.00    1.16 v _295_/A (sky130_fd_sc_hd__inv_2)
                  0.05    0.07    1.23 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                  0.05    0.00    1.23 ^ output10/A (sky130_fd_sc_hd__clkbuf_2)
                  0.25    0.23    1.46 ^ output10/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_yellow (net)
                  0.25    0.00    1.46 ^ a_yellow (out)
                                  1.46   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.46   data arrival time
---------------------------------------------------------------------------
                                 38.29   slack (MET)
```

ff_n40C_1v95

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.95 V.

Result: hold slack 0.04 ns; setup slack 38.54 ns - **MET**.

```
worst slack 38.54
worst slack 0.04
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.26    0.26   clock network delay (propagated)
                   0.14   0.00    0.26 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.03   0.21    0.47 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                   0.03   0.00    0.47 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                   0.03   0.07    0.53 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                   0.03   0.00    0.53 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.53   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.26    0.26   clock network delay (propagated)
                          0.25    0.51   clock uncertainty
                          0.00    0.51   clock reconvergence pessimism
                                  0.51 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.49   library hold time
                                  0.49   data required time
---------------------------------------------------------------------------
                                  0.49   data required time
                                 -0.53   data arrival time
---------------------------------------------------------------------------
                                  0.04   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_yellow (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.26    0.26   clock network delay (propagated)
                   0.09   0.00    0.26 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.03   0.21    0.47 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           state[3] (net)
                   0.03   0.00    0.47 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                   0.04   0.09    0.56 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                   0.04   0.00    0.56 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                   0.04   0.09    0.65 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                   0.04   0.00    0.65 v _209_/B (sky130_fd_sc_hd__or4_4)
                   0.06   0.31    0.96 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                   0.06   0.00    0.96 v _295_/A (sky130_fd_sc_hd__inv_2)
                   0.04   0.06    1.02 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                   0.04   0.00    1.02 ^ output10/A (sky130_fd_sc_hd__clkbuf_2)
                   0.21   0.19    1.21 ^ output10/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_yellow (net)
                   0.21   0.00    1.21 ^ a_yellow (out)
                                  1.21   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.21   data arrival time
---------------------------------------------------------------------------
                                 38.54   slack (MET)
```

ss_100C_1v40

Meaning: Slow NMOS / Slow PMOS; temperature +100 C; supply 1.40 V.

Result: hold slack 1.06 ns; setup slack 34.50 ns - **MET**.

```
worst slack 34.50
worst slack 1.06
Startpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          1.03    1.03   clock network delay (propagated)
                  0.33    0.00    1.03 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.07    0.94    1.97 v _364_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           cnt[7] (net)
                  0.07    0.00    1.97 v _227_/C1 (sky130_fd_sc_hd__o2111a_1)
                  0.07    0.30    2.28 v _227_/X (sky130_fd_sc_hd__o2111a_1)
     1    0.00                           _064_ (net)
                  0.07    0.00    2.28 v _364_/D (sky130_fd_sc_hd__dfxtp_1)
                                  2.28   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          1.03    1.03   clock network delay (propagated)
                          0.25    1.28   clock uncertainty
                          0.00    1.28   clock reconvergence pessimism
                                  1.28 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.07    1.22   library hold time
                                  1.22   data required time
---------------------------------------------------------------------------
                                  1.22   data required time
                                 -2.28   data arrival time
---------------------------------------------------------------------------
                                  1.06   slack (MET)
Startpoint: _339_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: ped_walk (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.97    0.97   clock network delay (propagated)
                  0.24    0.00    0.97 ^ _339_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.18    1.07    2.04 ^ _339_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01                           state[2] (net)
                  0.18    0.00    2.04 ^ _178_/A (sky130_fd_sc_hd__inv_2)
                  0.06    0.13    2.17 v _178_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _078_ (net)
                  0.06    0.00    2.17 v _181_/A (sky130_fd_sc_hd__or3_4)
                  0.19    1.14    3.32 v _181_/X (sky130_fd_sc_hd__or3_4)
     3    0.01                           _079_ (net)
                  0.19    0.00    3.32 v _208_/B (sky130_fd_sc_hd__or2_2)
                  0.22    1.08    4.40 v _208_/X (sky130_fd_sc_hd__or2_2)
     4    0.01                           _103_ (net)
                  0.22    0.00    4.40 v _293_/A (sky130_fd_sc_hd__inv_2)
                  0.14    0.22    4.61 ^ _293_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net14 (net)
                  0.14    0.00    4.61 ^ output14/A (sky130_fd_sc_hd__clkbuf_2)
                  0.58    0.63    5.25 ^ output14/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           ped_walk (net)
                  0.58    0.00    5.25 ^ ped_walk (out)
                                  5.25   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -5.25   data arrival time
---------------------------------------------------------------------------
                                 34.50   slack (MET)
```

ss_100C_1v60

Meaning: Slow NMOS / Slow PMOS; temperature +100 C; supply 1.60 V.

Result: hold slack 0.67 ns; setup slack 36.16 ns - **MET**.

```
worst slack 36.16
worst slack 0.67
Startpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.74    0.74   clock network delay (propagated)
                  0.24    0.00    0.74 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.06    0.66    1.40 v _364_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           cnt[7] (net)
                  0.06    0.00    1.40 v _227_/C1 (sky130_fd_sc_hd__o2111a_1)
                  0.05    0.21    1.60 v _227_/X (sky130_fd_sc_hd__o2111a_1)
     1    0.00                           _064_ (net)
                  0.05    0.00    1.60 v _364_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.60   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.74    0.74   clock network delay (propagated)
                          0.25    0.99   clock uncertainty
                          0.00    0.99   clock reconvergence pessimism
                                  0.99 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.06    0.93   library hold time
                                  0.93   data required time
---------------------------------------------------------------------------
                                  0.93   data required time
                                 -1.60   data arrival time
---------------------------------------------------------------------------
                                  0.67   slack (MET)
Startpoint: _339_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: ped_walk (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.69    0.69   clock network delay (propagated)
                  0.18    0.00    0.69 ^ _339_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.14    0.73    1.42 ^ _339_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01                           state[2] (net)
                  0.14    0.00    1.42 ^ _178_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.09    1.51 v _178_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _078_ (net)
                  0.04    0.00    1.51 v _181_/A (sky130_fd_sc_hd__or3_4)
                  0.13    0.75    2.27 v _181_/X (sky130_fd_sc_hd__or3_4)
     3    0.01                           _079_ (net)
                  0.13    0.00    2.27 v _208_/B (sky130_fd_sc_hd__or2_2)
                  0.16    0.71    2.98 v _208_/X (sky130_fd_sc_hd__or2_2)
     4    0.01                           _103_ (net)
                  0.16    0.00    2.98 v _293_/A (sky130_fd_sc_hd__inv_2)
                  0.10    0.15    3.13 ^ _293_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net14 (net)
                  0.10    0.00    3.13 ^ output14/A (sky130_fd_sc_hd__clkbuf_2)
                  0.43    0.46    3.59 ^ output14/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           ped_walk (net)
                  0.43    0.00    3.59 ^ ped_walk (out)
                                  3.59   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -3.59   data arrival time
---------------------------------------------------------------------------
                                 36.16   slack (MET)
```

ss_n40C_1v28

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.28 V.

Result: hold slack 2.34 ns; setup slack 7.35 ns - **MET**.

```
worst slack 7.35
worst slack 2.34
Startpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          2.37    2.37   clock network delay (propagated)
                  0.78    0.00    2.37 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.14    2.01    4.38 v _362_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           cnt[5] (net)
                  0.14    0.00    4.38 v _232_/A (sky130_fd_sc_hd__nand2_1)
                  0.23    0.29    4.67 ^ _232_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _124_ (net)
                  0.23    0.00    4.67 ^ _233_/A2 (sky130_fd_sc_hd__a211oi_1)
                  0.08    0.25    4.92 v _233_/Y (sky130_fd_sc_hd__a211oi_1)
     1    0.00                           _062_ (net)
                  0.08    0.00    4.92 v _362_/D (sky130_fd_sc_hd__dfxtp_1)
                                  4.92   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          2.37    2.37   clock network delay (propagated)
                          0.25    2.62   clock uncertainty
                          0.00    2.62   clock reconvergence pessimism
                                  2.62 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.03    2.58   library hold time
                                  2.58   data required time
---------------------------------------------------------------------------
                                  2.58   data required time
                                 -4.92   data arrival time
---------------------------------------------------------------------------
                                  2.34   slack (MET)
Startpoint: _357_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _350_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          2.37    2.37   clock network delay (propagated)
                  0.78    0.00    2.37 ^ _357_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.20    2.06    4.43 v _357_/Q (sky130_fd_sc_hd__dfxtp_1)
     4    0.01                           cnt[0] (net)
                  0.20    0.00    4.43 v _165_/B (sky130_fd_sc_hd__or2_1)
                  0.29    1.85    6.28 v _165_/X (sky130_fd_sc_hd__or2_1)
     3    0.01                           _066_ (net)
                  0.29    0.00    6.28 v _166_/B (sky130_fd_sc_hd__or2_1)
                  0.27    1.87    8.16 v _166_/X (sky130_fd_sc_hd__or2_1)
     3    0.00                           _067_ (net)
                  0.27    0.00    8.16 v _167_/B (sky130_fd_sc_hd__or2_2)
                  0.40    2.69   10.85 v _167_/X (sky130_fd_sc_hd__or2_2)
     5    0.01                           _068_ (net)
                  0.40    0.00   10.85 v _170_/D (sky130_fd_sc_hd__or4_4)
                  0.79    5.35   16.20 v _170_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _071_ (net)
                  0.79    0.00   16.20 v _171_/A (sky130_fd_sc_hd__inv_2)
                  0.18    0.67   16.87 ^ _171_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           _072_ (net)
                  0.18    0.00   16.87 ^ _172_/A (sky130_fd_sc_hd__buf_4)
                  0.64    0.81   17.68 ^ _172_/X (sky130_fd_sc_hd__buf_4)
    12    0.03                           _028_ (net)
                  0.64    0.00   17.68 ^ _323_/S (sky130_fd_sc_hd__mux2_1)
                  0.33    3.18   20.85 v _323_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _005_ (net)
                  0.33    0.00   20.85 v _218_/A3 (sky130_fd_sc_hd__o31a_1)
                  0.26    2.45   23.31 v _218_/X (sky130_fd_sc_hd__o31a_1)
     1    0.00                           _113_ (net)
                  0.26    0.00   23.31 v _219_/C1 (sky130_fd_sc_hd__o221a_1)
                  0.23    0.76   24.07 v _219_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _114_ (net)
                  0.23    0.00   24.07 v _220_/C1 (sky130_fd_sc_hd__o221a_1)
                  0.22    0.73   24.80 v _220_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _115_ (net)
                  0.22    0.00   24.80 v _221_/B1 (sky130_fd_sc_hd__o311a_2)
                  0.49    1.09   25.89 v _221_/X (sky130_fd_sc_hd__o311a_2)
     4    0.02                           _116_ (net)
                  0.49    0.00   25.89 v _223_/A2 (sky130_fd_sc_hd__o22a_1)
                  0.19    1.68   27.57 v _223_/X (sky130_fd_sc_hd__o22a_1)
     1    0.00                           _118_ (net)
                  0.19    0.00   27.57 v _224_/D (sky130_fd_sc_hd__or4_4)
                  0.77    5.14   32.71 v _224_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _119_ (net)
                  0.77    0.00   32.71 v _225_/A (sky130_fd_sc_hd__inv_2)
                  0.34    0.79   33.50 ^ _225_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _120_ (net)
                  0.34    0.00   33.50 ^ _226_/A (sky130_fd_sc_hd__clkbuf_4)
                  0.56    0.92   34.43 ^ _226_/X (sky130_fd_sc_hd__clkbuf_4)
    10    0.02                           _014_ (net)
                  0.56    0.00   34.43 ^ _336_/S (sky130_fd_sc_hd__mux2_1)
                  0.41    3.28   37.70 v _336_/X (sky130_fd_sc_hd__mux2_1)
     1    0.01                           _016_ (net)
                  0.41    0.00   37.70 v _244_/B1 (sky130_fd_sc_hd__a21oi_4)
                  0.94    1.18   38.88 ^ _244_/Y (sky130_fd_sc_hd__a21oi_4)
     5    0.01                           _130_ (net)
                  0.94    0.00   38.88 ^ _245_/A (sky130_fd_sc_hd__inv_2)
```

```
                0.15    0.54   39.42 v _245_/Y (sky130_fd_sc_hd__inv_2)
      3    0.01                       _131_ (net)
                0.15    0.00   39.42 v _272_/B (sky130_fd_sc_hd__or2_1)
                0.28    1.82   41.24 v _272_/X (sky130_fd_sc_hd__or2_1)
      2    0.01                       _152_ (net)
                0.28    0.00   41.24 v _274_/B1 (sky130_fd_sc_hd__o221a_1)
                0.22    2.02   43.26 v _274_/X (sky130_fd_sc_hd__o221a_1)
      1    0.00                       _050_ (net)
                0.22    0.00   43.26 v _350_/D (sky130_fd_sc_hd__dfxtp_1)
                               43.26   data arrival time
                       50.00   50.00   clock clk (rise edge)
                        1.96   51.96   clock network delay (propagated)
                       -0.25   51.71   clock uncertainty
                        0.00   51.71   clock reconvergence pessimism
                               51.71 ^ _350_/CLK (sky130_fd_sc_hd__dfxtp_1)
                       -1.09   50.62   library setup time
                               50.62   data required time
-----------------------------------------------------------------------------
                               50.62   data required time
                              -43.26   data arrival time
-----------------------------------------------------------------------------
                                7.35   slack (MET)
```

ss_n40C_1v35

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.35 V.

Result: hold slack 1.70 ns; setup slack 21.06 ns - **MET**.

```
worst slack 21.06
worst slack 1.70
Startpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          1.77    1.77   clock network delay (propagated)
                   0.61    0.00    1.77 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.11    1.50    3.28 v _362_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           cnt[5] (net)
                   0.11    0.00    3.28 v _232_/A (sky130_fd_sc_hd__nand2_1)
                   0.17    0.22    3.50 ^ _232_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _124_ (net)
                   0.17    0.00    3.50 ^ _233_/A2 (sky130_fd_sc_hd__a211oi_1)
                   0.07    0.19    3.69 v _233_/Y (sky130_fd_sc_hd__a211oi_1)
     1    0.00                           _062_ (net)
                   0.07    0.00    3.69 v _362_/D (sky130_fd_sc_hd__dfxtp_1)
                                   3.69   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          1.77    1.77   clock network delay (propagated)
                          0.25    2.02   clock uncertainty
                          0.00    2.02   clock reconvergence pessimism
                                  2.02 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.04    1.99   library hold time
                                  1.99   data required time
---------------------------------------------------------------------------
                                  1.99   data required time
                                 -3.69   data arrival time
---------------------------------------------------------------------------
                                  1.70   slack (MET)
Startpoint: _357_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _350_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          1.77    1.77   clock network delay (propagated)
                   0.61    0.00    1.77 ^ _357_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.16    1.55    3.32 v _357_/Q (sky130_fd_sc_hd__dfxtp_1)
     4    0.01                           cnt[0] (net)
                   0.16    0.00    3.32 v _165_/B (sky130_fd_sc_hd__or2_1)
                   0.22    1.29    4.62 v _165_/X (sky130_fd_sc_hd__or2_1)
     3    0.01                           _066_ (net)
                   0.22    0.00    4.62 v _166_/B (sky130_fd_sc_hd__or2_1)
                   0.20    1.30    5.92 v _166_/X (sky130_fd_sc_hd__or2_1)
     3    0.00                           _067_ (net)
                   0.20    0.00    5.92 v _167_/B (sky130_fd_sc_hd__or2_2)
                   0.28    1.85    7.77 v _167_/X (sky130_fd_sc_hd__or2_2)
     5    0.01                           _068_ (net)
                   0.28    0.00    7.77 v _170_/D (sky130_fd_sc_hd__or4_4)
                   0.51    3.44   11.21 v _170_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _071_ (net)
                   0.51    0.00   11.21 v _171_/A (sky130_fd_sc_hd__inv_2)
                   0.14    0.43   11.65 ^ _171_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           _072_ (net)
                   0.14    0.00   11.65 ^ _172_/A (sky130_fd_sc_hd__buf_4)
                   0.47    0.61   12.25 ^ _172_/X (sky130_fd_sc_hd__buf_4)
    12    0.03                           _028_ (net)
                   0.47    0.00   12.25 ^ _323_/S (sky130_fd_sc_hd__mux2_1)
                   0.23    2.12   14.37 v _323_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _005_ (net)
                   0.23    0.00   14.37 v _218_/A3 (sky130_fd_sc_hd__o31a_1)
                   0.18    1.60   15.98 v _218_/X (sky130_fd_sc_hd__o31a_1)
     1    0.00                           _113_ (net)
                   0.18    0.00   15.98 v _219_/C1 (sky130_fd_sc_hd__o221a_1)
                   0.16    0.54   16.52 v _219_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _114_ (net)
                   0.16    0.00   16.52 v _220_/C1 (sky130_fd_sc_hd__o221a_1)
                   0.15    0.51   17.03 v _220_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _115_ (net)
                   0.15    0.00   17.03 v _221_/B1 (sky130_fd_sc_hd__o311a_2)
                   0.34    0.76   17.79 v _221_/X (sky130_fd_sc_hd__o311a_2)
     4    0.02                           _116_ (net)
                   0.34    0.00   17.79 v _223_/A2 (sky130_fd_sc_hd__o22a_1)
                   0.13    1.14   18.93 v _223_/X (sky130_fd_sc_hd__o22a_1)
     1    0.00                           _118_ (net)
                   0.13    0.00   18.93 v _224_/D (sky130_fd_sc_hd__or4_4)
                   0.50    3.31   22.24 v _224_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _119_ (net)
                   0.50    0.00   22.24 v _225_/A (sky130_fd_sc_hd__inv_2)
                   0.26    0.53   22.77 ^ _225_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _120_ (net)
                   0.26    0.00   22.77 ^ _226_/A (sky130_fd_sc_hd__clkbuf_4)
                   0.40    0.70   23.47 ^ _226_/X (sky130_fd_sc_hd__clkbuf_4)
    10    0.03                           _014_ (net)
                   0.40    0.00   23.47 ^ _336_/S (sky130_fd_sc_hd__mux2_1)
                   0.29    2.20   25.67 v _336_/X (sky130_fd_sc_hd__mux2_1)
     1    0.01                           _016_ (net)
                   0.29    0.00   25.67 v _244_/B1 (sky130_fd_sc_hd__a21oi_4)
                   0.65    0.82   26.49 ^ _244_/Y (sky130_fd_sc_hd__a21oi_4)
     5    0.01                           _130_ (net)
                   0.65    0.00   26.49 ^ _245_/A (sky130_fd_sc_hd__inv_2)
```

```
                0.11    0.36   26.85 v _245_/Y (sky130_fd_sc_hd__inv_2)
     3    0.01                         _131_ (net)
                0.11    0.00   26.85 v _272_/B (sky130_fd_sc_hd__or2_1)
                0.21    1.26   28.11 v _272_/X (sky130_fd_sc_hd__or2_1)
     2    0.01                         _152_ (net)
                0.21    0.00   28.11 v _274_/B1 (sky130_fd_sc_hd__o221a_1)
                0.15    1.34   29.45 v _274_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                         _050_ (net)
                0.15    0.00   29.45 v _350_/D (sky130_fd_sc_hd__dfxtp_1)
                               29.45   data arrival time
                       50.00   50.00   clock clk (rise edge)
                        1.50   51.50   clock network delay (propagated)
                       -0.25   51.25   clock uncertainty
                        0.00   51.25   clock reconvergence pessimism
                               51.25 ^ _350_/CLK (sky130_fd_sc_hd__dfxtp_1)
                       -0.74   50.51   library setup time
                               50.51   data required time
-----------------------------------------------------------------------------
                               50.51   data required time
                              -29.45   data arrival time
-----------------------------------------------------------------------------
                               21.06   slack (MET)
```

ss_n40C_1v40

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.40 V.

Result: hold slack 1.39 ns; setup slack 26.88 ns - **MET**.

```
worst slack 26.88
worst slack 1.39
Startpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          1.49    1.49   clock network delay (propagated)
                   0.51   0.00    1.49 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.10   1.25    2.74 v _362_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           cnt[5] (net)
                   0.10   0.00    2.74 v _232_/A (sky130_fd_sc_hd__nand2_1)
                   0.14   0.19    2.93 ^ _232_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _124_ (net)
                   0.14   0.00    2.93 ^ _233_/A2 (sky130_fd_sc_hd__a211oi_1)
                   0.06   0.16    3.09 v _233_/Y (sky130_fd_sc_hd__a211oi_1)
     1    0.00                           _062_ (net)
                   0.06   0.00    3.09 v _362_/D (sky130_fd_sc_hd__dfxtp_1)
                                  3.09   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          1.49    1.49   clock network delay (propagated)
                          0.25    1.74   clock uncertainty
                          0.00    1.74   clock reconvergence pessimism
                                  1.74 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.04    1.70   library hold time
                                  1.70   data required time
---------------------------------------------------------------------------
                                  1.70   data required time
                                 -3.09   data arrival time
---------------------------------------------------------------------------
                                  1.39   slack (MET)
Startpoint: _357_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _350_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          1.49    1.49   clock network delay (propagated)
                   0.51   0.00    1.49 ^ _357_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.14   1.30    2.78 v _357_/Q (sky130_fd_sc_hd__dfxtp_1)
     4    0.01                           cnt[0] (net)
                   0.14   0.00    2.78 v _165_/B (sky130_fd_sc_hd__or2_1)
                   0.18   1.04    3.83 v _165_/X (sky130_fd_sc_hd__or2_1)
     3    0.01                           _066_ (net)
                   0.18   0.00    3.83 v _166_/B (sky130_fd_sc_hd__or2_1)
                   0.17   1.05    4.88 v _166_/X (sky130_fd_sc_hd__or2_1)
     3    0.00                           _067_ (net)
                   0.17   0.00    4.88 v _167_/B (sky130_fd_sc_hd__or2_2)
                   0.23   1.48    6.36 v _167_/X (sky130_fd_sc_hd__or2_2)
     5    0.01                           _068_ (net)
                   0.23   0.00    6.36 v _170_/D (sky130_fd_sc_hd__or4_4)
                   0.40   2.67    9.03 v _170_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _071_ (net)
                   0.40   0.00    9.03 v _171_/A (sky130_fd_sc_hd__inv_2)
                   0.11   0.34    9.37 ^ _171_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           _072_ (net)
                   0.11   0.00    9.37 ^ _172_/A (sky130_fd_sc_hd__buf_4)
                   0.38   0.51    9.88 ^ _172_/X (sky130_fd_sc_hd__buf_4)
    12    0.03                           _028_ (net)
                   0.38   0.00    9.88 ^ _323_/S (sky130_fd_sc_hd__mux2_1)
                   0.18   1.68   11.56 v _323_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _005_ (net)
                   0.18   0.00   11.56 v _218_/A3 (sky130_fd_sc_hd__o31a_1)
                   0.15   1.25   12.81 v _218_/X (sky130_fd_sc_hd__o31a_1)
     1    0.00                           _113_ (net)
                   0.15   0.00   12.81 v _219_/C1 (sky130_fd_sc_hd__o221a_1)
                   0.13   0.44   13.25 v _219_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _114_ (net)
                   0.13   0.00   13.25 v _220_/C1 (sky130_fd_sc_hd__o221a_1)
                   0.12   0.42   13.67 v _220_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _115_ (net)
                   0.12   0.00   13.67 v _221_/B1 (sky130_fd_sc_hd__o311a_2)
                   0.28   0.62   14.29 v _221_/X (sky130_fd_sc_hd__o311a_2)
     4    0.02                           _116_ (net)
                   0.28   0.00   14.29 v _223_/A2 (sky130_fd_sc_hd__o22a_1)
                   0.11   0.92   15.21 v _223_/X (sky130_fd_sc_hd__o22a_1)
     1    0.00                           _118_ (net)
                   0.11   0.00   15.21 v _224_/D (sky130_fd_sc_hd__or4_4)
                   0.39   2.57   17.78 v _224_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _119_ (net)
                   0.39   0.00   17.78 v _225_/A (sky130_fd_sc_hd__inv_2)
                   0.22   0.42   18.20 ^ _225_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _120_ (net)
                   0.22   0.00   18.20 ^ _226_/A (sky130_fd_sc_hd__clkbuf_4)
                   0.33   0.59   18.79 ^ _226_/X (sky130_fd_sc_hd__clkbuf_4)
    10    0.03                           _014_ (net)
                   0.33   0.00   18.79 ^ _336_/S (sky130_fd_sc_hd__mux2_1)
                   0.24   1.75   20.54 v _336_/X (sky130_fd_sc_hd__mux2_1)
     1    0.01                           _016_ (net)
                   0.24   0.00   20.54 v _244_/B1 (sky130_fd_sc_hd__a21oi_4)
                   0.53   0.67   21.20 ^ _244_/Y (sky130_fd_sc_hd__a21oi_4)
     5    0.01                           _130_ (net)
                   0.53   0.00   21.20 ^ _245_/A (sky130_fd_sc_hd__inv_2)
```

```
                0.09    0.29   21.49 v _245_/Y (sky130_fd_sc_hd__inv_2)
     3    0.01                         _131_ (net)
                0.09    0.00   21.49 v _272_/B (sky130_fd_sc_hd__or2_1)
                0.18    1.01   22.51 v _272_/X (sky130_fd_sc_hd__or2_1)
     2    0.01                         _152_ (net)
                0.18    0.00   22.51 v _274_/B1 (sky130_fd_sc_hd__o221a_1)
                0.13    1.06   23.57 v _274_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                         _050_ (net)
                0.13    0.00   23.57 v _350_/D (sky130_fd_sc_hd__dfxtp_1)
                               23.57   data arrival time
                       50.00   50.00   clock clk (rise edge)
                        1.27   51.27   clock network delay (propagated)
                       -0.25   51.02   clock uncertainty
                        0.00   51.02   clock reconvergence pessimism
                               51.02 ^ _350_/CLK (sky130_fd_sc_hd__dfxtp_1)
                       -0.57   50.45   library setup time
                               50.45   data required time
-----------------------------------------------------------------------------
                               50.45   data required time
                              -23.57   data arrival time
-----------------------------------------------------------------------------
                               26.88   slack (MET)
```

ss_n40C_1v44

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.44 V.

Result: hold slack 1.20 ns; setup slack 30.28 ns - **MET**.

```
worst slack 30.28
worst slack 1.20
Startpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _362_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          1.29    1.29   clock network delay (propagated)
                   0.43   0.00    1.29 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.09   1.09    2.39 v _362_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           cnt[5] (net)
                   0.09   0.00    2.39 v _232_/A (sky130_fd_sc_hd__nand2_1)
                   0.12   0.17    2.56 ^ _232_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _124_ (net)
                   0.12   0.00    2.56 ^ _233_/A2 (sky130_fd_sc_hd__a211oi_1)
                   0.06   0.15    2.70 v _233_/Y (sky130_fd_sc_hd__a211oi_1)
     1    0.00                           _062_ (net)
                   0.06   0.00    2.70 v _362_/D (sky130_fd_sc_hd__dfxtp_1)
                                  2.70   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          1.29    1.29   clock network delay (propagated)
                          0.25    1.54   clock uncertainty
                          0.00    1.54   clock reconvergence pessimism
                                  1.54 ^ _362_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.04    1.51   library hold time
                                  1.51   data required time
---------------------------------------------------------------------------
                                  1.51   data required time
                                 -2.70   data arrival time
---------------------------------------------------------------------------
                                  1.20   slack (MET)
Startpoint: _357_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _350_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          1.29    1.29   clock network delay (propagated)
                   0.43   0.00    1.29 ^ _357_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.13   1.13    2.43 v _357_/Q (sky130_fd_sc_hd__dfxtp_1)
     4    0.01                           cnt[0] (net)
                   0.13   0.00    2.43 v _165_/B (sky130_fd_sc_hd__or2_1)
                   0.16   0.90    3.32 v _165_/X (sky130_fd_sc_hd__or2_1)
     3    0.01                           _066_ (net)
                   0.16   0.00    3.32 v _166_/B (sky130_fd_sc_hd__or2_1)
                   0.15   0.90    4.22 v _166_/X (sky130_fd_sc_hd__or2_1)
     3    0.00                           _067_ (net)
                   0.15   0.00    4.22 v _167_/B (sky130_fd_sc_hd__or2_2)
                   0.20   1.27    5.49 v _167_/X (sky130_fd_sc_hd__or2_2)
     5    0.01                           _068_ (net)
                   0.20   0.00    5.49 v _170_/D (sky130_fd_sc_hd__or4_4)
                   0.34   2.24    7.73 v _170_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _071_ (net)
                   0.34   0.00    7.73 v _171_/A (sky130_fd_sc_hd__inv_2)
                   0.10   0.29    8.02 ^ _171_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           _072_ (net)
                   0.10   0.00    8.02 ^ _172_/A (sky130_fd_sc_hd__buf_4)
                   0.33   0.45    8.47 ^ _172_/X (sky130_fd_sc_hd__buf_4)
    12    0.03                           _028_ (net)
                   0.33   0.00    8.47 ^ _323_/S (sky130_fd_sc_hd__mux2_1)
                   0.16   1.42    9.89 v _323_/X (sky130_fd_sc_hd__mux2_1)
     1    0.00                           _005_ (net)
                   0.16   0.00    9.89 v _218_/A3 (sky130_fd_sc_hd__o31a_1)
                   0.13   1.06   10.95 v _218_/X (sky130_fd_sc_hd__o31a_1)
     1    0.00                           _113_ (net)
                   0.13   0.00   10.95 v _219_/C1 (sky130_fd_sc_hd__o221a_1)
                   0.12   0.38   11.33 v _219_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _114_ (net)
                   0.12   0.00   11.33 v _220_/C1 (sky130_fd_sc_hd__o221a_1)
                   0.11   0.37   11.69 v _220_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                           _115_ (net)
                   0.11   0.00   11.69 v _221_/B1 (sky130_fd_sc_hd__o311a_2)
                   0.24   0.54   12.23 v _221_/X (sky130_fd_sc_hd__o311a_2)
     4    0.02                           _116_ (net)
                   0.24   0.00   12.23 v _223_/A2 (sky130_fd_sc_hd__o22a_1)
                   0.10   0.78   13.02 v _223_/X (sky130_fd_sc_hd__o22a_1)
     1    0.00                           _118_ (net)
                   0.10   0.00   13.02 v _224_/D (sky130_fd_sc_hd__or4_4)
                   0.33   2.15   15.16 v _224_/X (sky130_fd_sc_hd__or4_4)
     4    0.01                           _119_ (net)
                   0.33   0.00   15.16 v _225_/A (sky130_fd_sc_hd__inv_2)
                   0.19   0.36   15.52 ^ _225_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _120_ (net)
                   0.19   0.00   15.53 ^ _226_/A (sky130_fd_sc_hd__clkbuf_4)
                   0.29   0.52   16.04 ^ _226_/X (sky130_fd_sc_hd__clkbuf_4)
    10    0.03                           _014_ (net)
                   0.29   0.00   16.04 ^ _336_/S (sky130_fd_sc_hd__mux2_1)
                   0.21   1.49   17.53 v _336_/X (sky130_fd_sc_hd__mux2_1)
     1    0.01                           _016_ (net)
                   0.21   0.00   17.53 v _244_/B1 (sky130_fd_sc_hd__a21oi_4)
                   0.46   0.57   18.10 ^ _244_/Y (sky130_fd_sc_hd__a21oi_4)
     5    0.01                           _130_ (net)
                   0.46   0.00   18.10 ^ _245_/A (sky130_fd_sc_hd__inv_2)
```

```
                0.08    0.25   18.35 v _245_/Y (sky130_fd_sc_hd__inv_2)
     3    0.01                        _131_ (net)
                0.08    0.00   18.35 v _272_/B (sky130_fd_sc_hd__or2_1)
                0.16    0.87   19.22 v _272_/X (sky130_fd_sc_hd__or2_1)
     2    0.01                        _152_ (net)
                0.16    0.00   19.22 v _274_/B1 (sky130_fd_sc_hd__o221a_1)
                0.11    0.90   20.12 v _274_/X (sky130_fd_sc_hd__o221a_1)
     1    0.00                        _050_ (net)
                0.11    0.00   20.12 v _350_/D (sky130_fd_sc_hd__dfxtp_1)
                               20.12   data arrival time
                       50.00   50.00   clock clk (rise edge)
                        1.10   51.10   clock network delay (propagated)
                       -0.25   50.85   clock uncertainty
                        0.00   50.85   clock reconvergence pessimism
                               50.85 ^ _350_/CLK (sky130_fd_sc_hd__dfxtp_1)
                       -0.45   50.40   library setup time
                               50.40   data required time
-----------------------------------------------------------------------------
                               50.40   data required time
                              -20.12   data arrival time
-----------------------------------------------------------------------------
                               30.28   slack (MET)
```

ss_n40C_1v60

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.60 V.

Result: hold slack 0.73 ns; setup slack 35.45 ns - **MET**.

```
worst slack 35.45
worst slack 0.73
Startpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.87    0.87   clock network delay (propagated)
                  0.31    0.00    0.87 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.71    1.58 v _364_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           cnt[7] (net)
                  0.05    0.00    1.58 v _227_/C1 (sky130_fd_sc_hd__o2111a_1)
                  0.04    0.24    1.82 v _227_/X (sky130_fd_sc_hd__o2111a_1)
     1    0.00                           _064_ (net)
                  0.04    0.00    1.82 v _364_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.82   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.87    0.87   clock network delay (propagated)
                          0.25    1.12   clock uncertainty
                          0.00    1.12   clock reconvergence pessimism
                                  1.12 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.04    1.09   library hold time
                                  1.09   data required time
---------------------------------------------------------------------------
                                  1.09   data required time
                                 -1.82   data arrival time
---------------------------------------------------------------------------
                                  0.73   slack (MET)
Startpoint: _339_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: ped_walk (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.81    0.81   clock network delay (propagated)
                  0.23    0.00    0.81 ^ _339_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.17    0.82    1.64 ^ _339_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01                           state[2] (net)
                  0.17    0.00    1.64 ^ _178_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.10    1.74 v _178_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _078_ (net)
                  0.04    0.00    1.74 v _181_/A (sky130_fd_sc_hd__or3_4)
                  0.15    1.02    2.75 v _181_/X (sky130_fd_sc_hd__or3_4)
     3    0.01                           _079_ (net)
                  0.15    0.00    2.75 v _208_/B (sky130_fd_sc_hd__or2_2)
                  0.14    0.80    3.55 v _208_/X (sky130_fd_sc_hd__or2_2)
     4    0.01                           _103_ (net)
                  0.14    0.00    3.55 v _293_/A (sky130_fd_sc_hd__inv_2)
                  0.12    0.18    3.74 ^ _293_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net14 (net)
                  0.12    0.00    3.74 ^ output14/A (sky130_fd_sc_hd__clkbuf_2)
                  0.56    0.56    4.30 ^ output14/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           ped_walk (net)
                  0.56    0.00    4.30 ^ ped_walk (out)
                                  4.30   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -4.30   data arrival time
---------------------------------------------------------------------------
                                 35.45   slack (MET)
```

ss_n40C_1v76

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.76 V.

Result: hold slack 0.47 ns; setup slack 36.65 ns - **MET**.

```
worst slack 36.65
worst slack 0.47
Startpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _364_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.66    0.66   clock network delay (propagated)
                   0.24   0.00    0.66 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.04   0.53    1.19 v _364_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           cnt[7] (net)
                   0.04   0.00    1.19 v _227_/C1 (sky130_fd_sc_hd__o2111a_1)
                   0.03   0.17    1.36 v _227_/X (sky130_fd_sc_hd__o2111a_1)
     1    0.00                           _064_ (net)
                   0.03   0.00    1.36 v _364_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.36   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.66    0.66   clock network delay (propagated)
                          0.25    0.91   clock uncertainty
                          0.00    0.91   clock reconvergence pessimism
                                  0.91 ^ _364_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02    0.89   library hold time
                                  0.89   data required time
---------------------------------------------------------------------------
                                  0.89   data required time
                                 -1.36   data arrival time
---------------------------------------------------------------------------
                                  0.47   slack (MET)
Startpoint: _339_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: ped_walk (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.61    0.61   clock network delay (propagated)
                   0.18   0.00    0.61 ^ _339_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   0.13   0.60    1.21 ^ _339_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01                           state[2] (net)
                   0.13   0.00    1.21 ^ _178_/A (sky130_fd_sc_hd__inv_2)
                   0.03   0.07    1.28 v _178_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _078_ (net)
                   0.03   0.00    1.28 v _181_/A (sky130_fd_sc_hd__or3_4)
                   0.11   0.70    1.98 v _181_/X (sky130_fd_sc_hd__or3_4)
     3    0.01                           _079_ (net)
                   0.11   0.00    1.98 v _208_/B (sky130_fd_sc_hd__or2_2)
                   0.11   0.56    2.54 v _208_/X (sky130_fd_sc_hd__or2_2)
     4    0.01                           _103_ (net)
                   0.11   0.00    2.54 v _293_/A (sky130_fd_sc_hd__inv_2)
                   0.09   0.13    2.67 ^ _293_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net14 (net)
                   0.09   0.00    2.67 ^ output14/A (sky130_fd_sc_hd__clkbuf_2)
                   0.42   0.42    3.10 ^ output14/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           ped_walk (net)
                   0.42   0.00    3.10 ^ ped_walk (out)
                                  3.10   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -3.10   data arrival time
---------------------------------------------------------------------------
                                 36.65   slack (MET)
```

tt_025C_1v80

Meaning: Typical NMOS / Typical PMOS; temperature +25 C; supply 1.80 V.

Result: hold slack 0.21 ns; setup slack 37.89 ns - **MET**.

```
worst slack 37.89
worst slack 0.21
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.37    0.37   clock network delay (propagated)
                  0.19    0.00    0.37 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.34    0.71 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.04    0.00    0.71 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                  0.04    0.10    0.82 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                  0.04    0.00    0.82 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.82   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.37    0.37   clock network delay (propagated)
                          0.25    0.62   clock uncertainty
                          0.00    0.62   clock reconvergence pessimism
                                  0.62 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.61   library hold time
                                  0.61   data required time
---------------------------------------------------------------------------
                                  0.61   data required time
                                 -0.82   data arrival time
---------------------------------------------------------------------------
                                  0.21   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.39    0.39   clock network delay (propagated)
                  0.12    0.00    0.39 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.33    0.72 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           state[3] (net)
                  0.04    0.00    0.72 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.06    0.13    0.84 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                  0.06    0.00    0.84 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.06    0.14    0.98 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                  0.06    0.00    0.98 v _209_/B (sky130_fd_sc_hd__or4_4)
                  0.09    0.53    1.51 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                  0.09    0.00    1.51 v _295_/A (sky130_fd_sc_hd__inv_2)
                  0.05    0.08    1.59 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                  0.05    0.00    1.59 ^ _297_/B (sky130_fd_sc_hd__nor2_1)
                  0.04    0.05    1.64 v _297_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.01                           net9 (net)
                  0.04    0.00    1.64 v output9/A (sky130_fd_sc_hd__clkbuf_2)
                  0.19    0.23    1.86 v output9/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_red (net)
                  0.19    0.00    1.86 v a_red (out)
                                  1.86   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.86   data arrival time
---------------------------------------------------------------------------
                                 37.89   slack (MET)
```

tt_100C_1v80

Meaning: Typical NMOS / Typical PMOS; temperature +100 C; supply 1.80 V.

Result: hold slack 0.21 ns; setup slack 37.89 ns - **MET**.

```
worst slack 37.89
worst slack 0.21
Startpoint: _341_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _342_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.38    0.38   clock network delay (propagated)
                  0.19    0.00    0.38 ^ _341_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.34    0.72 ^ _341_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.04    0.00    0.72 ^ _288_/B (sky130_fd_sc_hd__and2_1)
                  0.04    0.11    0.83 ^ _288_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _042_ (net)
                  0.04    0.00    0.83 ^ _342_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.83   data arrival time
                          0.00    0.00   clock clk (rise edge)
                          0.38    0.38   clock network delay (propagated)
                          0.25    0.63   clock uncertainty
                          0.00    0.63   clock reconvergence pessimism
                                  0.63 ^ _342_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.61   library hold time
                                  0.61   data required time
---------------------------------------------------------------------------
                                  0.61   data required time
                                 -0.83   data arrival time
---------------------------------------------------------------------------
                                  0.21   slack (MET)
Startpoint: _340_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: a_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.39    0.39   clock network delay (propagated)
                  0.12    0.00    0.39 ^ _340_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.33    0.73 v _340_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.01                           state[3] (net)
                  0.04    0.00    0.73 v _176_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.07    0.13    0.86 v _176_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _076_ (net)
                  0.07    0.00    0.86 v _177_/A (sky130_fd_sc_hd__clkbuf_2)
                  0.07    0.14    1.00 v _177_/X (sky130_fd_sc_hd__clkbuf_2)
     5    0.01                           _077_ (net)
                  0.07    0.00    1.00 v _209_/B (sky130_fd_sc_hd__or4_4)
                  0.09    0.50    1.49 v _209_/X (sky130_fd_sc_hd__or4_4)
     3    0.01                           _104_ (net)
                  0.09    0.00    1.49 v _295_/A (sky130_fd_sc_hd__inv_2)
                  0.05    0.07    1.57 ^ _295_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net10 (net)
                  0.05    0.00    1.57 ^ _297_/B (sky130_fd_sc_hd__nor2_1)
                  0.04    0.05    1.62 v _297_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.01                           net9 (net)
                  0.04    0.00    1.62 v output9/A (sky130_fd_sc_hd__clkbuf_2)
                  0.21    0.24    1.86 v output9/X (sky130_fd_sc_hd__clkbuf_2)
     1    0.05                           a_red (net)
                  0.21    0.00    1.86 v a_red (out)
                                  1.86   data arrival time
                         50.00   50.00   clock clk (rise edge)
                          0.00   50.00   clock network delay (propagated)
                         -0.25   49.75   clock uncertainty
                          0.00   49.75   clock reconvergence pessimism
                        -10.00   39.75   output external delay
                                 39.75   data required time
---------------------------------------------------------------------------
                                 39.75   data required time
                                 -1.86   data arrival time
---------------------------------------------------------------------------
                                 37.89   slack (MET)
```

Conclusion

All 16 SKY130 HD corners were analyzed on the routed design with SPEF parasitics at a 50 ns clock. Worst setup slack is 7.35 ns (ss_n40C_1v28); worst hold slack is 0.04 ns (ff_n40C_1v95). Timing is met in all 16 corners.